

Mixed-Precision Amortized Defect Correction for Adams–Bashforth–Crank–Nicolson Schemes in Quasi-Geostrophic Vorticity Flows

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Abstract

Adams–Bashforth–Crank–Nicolson schemes are attractive for pseudo-spectral geophysical-flow solvers because they reuse the nonlinear history and require only one new advective evaluation per time step. Their efficiency is nevertheless restricted by the accuracy and stability limits of the explicit Adams–Bashforth update. We develop a mixed-precision amortized defect correction that maps a coarse AB2–CN2 step to a fine-step RK4 reference trajectory for two-dimensional quasi-geostrophic vorticity flows. The correction is obtained from the difference between the AB2–CN2 and RK4 local truncation errors. The AB2–CN2 defect is expanded through $\mathcal{O}(\Delta T^6)$ and separated into a Fourier-diagonal analytical contribution and a nonlinear contribution involving temporal derivatives of the advective Jacobian. A compact, physics-initialized network estimates only the nonlinear contribution from an S -level history stencil. Since quasi-geostrophic advection is quadratic, the estimator takes the form of an S -stage bilinear convolution approximation of the required higher-order derivatives. This structure yields an explicit error decomposition in which the temporal finite-difference error is driven by a single unresolved derivative, while a Wiener analysis characterizes the irreducible approximation floor and its dependence on the local decorrelation rate. The network is evaluated in Float32, whereas the prognostic state, spectral operators, nonlinear evaluation, correction assembly, and implicit solve remain in Float64. A modewise amplification analysis then propagates the implemented estimator error directly through the corrected scheme. The comparative protocol includes uncorrected and higher-history Adams–Bashforth–Crank–Nicolson schemes, the RK4 target, and a WENO-based end-to-end configuration under matched accuracy and computational-work criteria. **Key words.** amortized defect correction, mixed precision, AB2–CN2, Adams–Bashforth schemes, Wiener filtering, von Neumann stability, pseudo-spectral methods, quasi-geostrophic turbulence, scientific machine learning

1. Introduction. Adams–Bashforth–Crank–Nicolson discretizations combine an inexpensive explicit treatment of nonlinear transport with an implicit treatment of stiff linear dynamics [1, 11]. In a pseudo-spectral solver, the Crank–Nicolson operator is diagonal in Fourier space, while the Adams–Bashforth update reuses previously computed nonlinear terms. AB2–CN2 therefore requires only one new advective evaluation per step and little additional storage. This efficiency is valuable in quasi-geostrophic turbulence simulations, where the pseudo-spectral Jacobian dominates

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the per-step cost, but the usable coarse step remains limited by the order and stability of the explicit update.

We address this limitation by learning the defect between the inexpensive AB2–CN2 update and a fine-step RK4 reference. If $\Psi_{\Delta T}^{\text{AB2CN2}}$ denotes the coarse update and $\Phi_{\Delta T}^{\text{RK4}}$ denotes the RK4 flow accumulated over the same coarse interval, then the desired correction is $\Phi_{\Delta T}^{\text{RK4}} - \Psi_{\Delta T}^{\text{AB2CN2}}$. Its Taylor expansion is the difference between the two schemes’ local truncation errors. This identity provides the analytical structure of the correction and separates the contribution of the coarse Adams–Bashforth formula from that of the RK4 target.

The resulting operators contain powers of the Fourier-diagonal linear operator and temporal derivatives of the nonlinear Jacobian. The former are evaluated exactly at negligible marginal cost, whereas the latter require nested applications of the bilinear quasi-geostrophic advection operator. We therefore reconstruct only these nonlinear derivatives from an S -level history. A fixed Vandermonde layer forms the temporal finite differences, and a shallow bilinear convolution approximates the associated spatial derivative combinations. Because the architecture follows the quadratic structure of the governing equation, its error can be propagated analytically. The leading finite-difference errors share a single unresolved derivative, and the corresponding regression admits a Wiener decomposition [38] that identifies both the predictable correction and the irreducible floor of the local convolution class.

The mixed-precision implementation confines Float32 arithmetic to the network weights, activations, and predicted nonlinear-derivative channels. The prognostic state, spectral transforms, linear operators, nonlinear Jacobian, defect assembly, and Crank–Nicolson solve remain in Float64. The numerical backbone consequently retains double-precision arithmetic, while the learned component exploits lower-precision throughput. The complete corrected update is analyzed through its modewise amplification polynomial, with the implemented derivative-reconstruction error entering directly into the polynomial coefficients.

The principal contributions are fourfold. We first derive the AB2–CN2 local truncation operators through $\mathcal{O}(\Delta T^6)$ and construct the finite-order correction that matches the accumulated RK4 reference while preserving the single-evaluation structure of the coarse solver. We then derive the exact propagation of the temporal reconstruction error and characterize the Wiener floor of the bilinear convolution estimator. Next, we obtain the corrected modewise amplification polynomial and its stability criterion. Finally, we define comparisons with uncorrected AB2–CN2, higher-history Adams–Bashforth–Crank–Nicolson formulas, the RK4 target, and a WENO-based end-to-end discretization under matched accuracy and computational-work criteria.

Notation. Table 1 collects recurring symbols.

Outline. Section 2 introduces the quasi-geostrophic vorticity equation and its exact derivative recursions. Section 3 defines the AB2–CN2 backbone and the comparison formulas. Section 4 develops the error, cost, and stability analysis. Section 5 derives the amortized defect correction and the full numerical scheme. Section 6 presents the learned estimator and its Wiener error analysis. Section 7 propagates the implemented estimator error through the corrected amplification polynomial. Section 8 specifies the mixed-precision implementation and comparative evaluation protocol.

2. Governing equations.

Table 1: Key notation.

symbol	meaning
ω, ψ	vorticity and streamfunction, $\psi = \Delta^{-1}\omega$
$\mathcal{L}$	Fourier-diagonal linear operator
$N(\omega)$	nonlinear term $-\mathcal{J}(\psi, \omega) + F$
S	number of stored temporal levels
ΔT	coarse time step
τ	local truncation error
$\Delta T^*, \tau_\lambda$	convergence radius and nonlinear Taylor microscale
$\sigma(\kappa)$	per-shell decorrelation rate
z_L, z_N, z	modal linear, nonlinear, and combined symbols
$g(z)$	principal amplification factor
f_{NN}	learned nonlinear-derivative contribution

2.1. The QG vorticity equation. We consider the two-dimensional quasi-geostrophic vorticity equation on a doubly-periodic domain,

$$\partial_t \omega = \mathcal{L} \omega + N(\omega), \quad N(\omega) = -\mathcal{J}(\psi, \omega) + F, \quad \psi = \Delta^{-1} \omega, \quad \dot{F} = 0, \quad (1)$$

with $\mathcal{L} = \nu \Delta - \mu - \beta \partial_x \Delta^{-1}$ collecting viscosity, linear drag, and the β -plane term, Fourier symbol

$$\hat{\mathcal{L}}(\mathbf{k}) = -\nu \kappa^2 - \mu + i\beta k_x / \kappa^2, \quad \kappa = |\mathbf{k}|, \quad (2)$$

and $\mathcal{J}(a, b) = \partial_x a \partial_y b - \partial_y a \partial_x b$. Spatial discretization is pseudo-spectral with 2/3 dealiasing of the quadratic products [3, 6]; the physical setting follows Suresh Babu et al. [31]. For the one-step analysis it is convenient to write (1) as an autonomous ODE,

$$\dot{\omega} = G(\omega) = \mathcal{L} \omega + B(\omega, \omega) + F, \quad B(a, b) = -\frac{1}{2} [\mathcal{J}(\Delta^{-1} a, b) + \mathcal{J}(\Delta^{-1} b, a)], \quad (3)$$

with B symmetric bilinear and $B(\omega, \omega) = -\mathcal{J}(\Delta^{-1} \omega, \omega)$, so $N = B(\omega, \omega) + F$.

2.2. Exact derivative recursions. Repeated differentiation of (1) along the trajectory gives the mutually recursive relations used throughout:

$$\omega^{(k)} = \mathcal{L} \omega^{(k-1)} + N^{(k-1)}, \quad \psi^{(k)} = \Delta^{-1} \omega^{(k)}, \quad (4)$$

$$N^{(m)} = -\sum_{j=0}^m \binom{m}{j} \mathcal{J}(\psi^{(m-j)}, \omega^{(j)}), \quad m \geq 1, \quad (5)$$

where the static forcing drops out of every $N^{(m)}$, $m \geq 1$ (bilinearity of $\mathcal{J}$ and the Leibniz rule). Unrolled,

$$\omega^{(k)} = \mathcal{L}^k \omega + \sum_{j=0}^{k-1} \mathcal{L}^{k-1-j} N^{(j)}, \quad \text{e.g.} \quad \dot{N} = -\mathcal{J}(\dot{\psi}, \omega) - \mathcal{J}(\psi, \dot{\omega}). \quad (6)$$

Each additional time derivative of N costs another layer of nested Jacobians; this nested-Jacobian cost is what the closure will learn to avoid. Finally, because G is quadratic in ω (B is bilinear), its Taylor–Fréchet expansion about any state has exactly three terms, the value, the Jacobian, and a constant second derivative, and every higher derivative vanishes identically:

$$G(\omega + p) = f + Jp + B(p, p), \quad f \equiv G(\omega), \quad J \equiv G'(\omega) = \mathcal{L} + 2B(\omega, \cdot), \quad G''' \equiv 0. \quad (7)$$

3. Time-integration schemes.

3.1. General forms. An s -step Adams–Bashforth formula for $\dot{\omega} = G(\omega)$ is written as

$$\sum_{j=0}^s \alpha_j \omega^{n+j} = h \sum_{j=0}^s \beta_j G(\omega^{n+j}), \quad \alpha_s = 1, \quad (8)$$

explicit when $\beta_s = 0$. With the generating polynomials $\rho(\zeta) = \sum_j \alpha_j \zeta^j$ and $\sigma(\zeta) = \sum_j \beta_j \zeta^j$, the method has order p if and only if

$$\rho(e^h) - h \sigma(e^h) = \mathcal{O}(h^{p+1}), \quad \text{equivalently} \quad C_q \equiv \sum_{j=0}^s \left[\alpha_j \frac{j^q}{q!} - \beta_j \frac{j^{q-1}}{(q-1)!} \right] = 0, \quad q = 0, \dots, p, \quad (9)$$

and the first nonzero constant C_{p+1} sets the leading LTE [18, Sec. III.2]. For the split right-hand side $G = \mathcal{L}\omega + N$ of (1), the Adams–Bashforth–Crank–Nicolson family [1, 11] applies the Crank–Nicolson polynomial to the stiff linear part and the Adams–Bashforth polynomial to the nonlinearity:

$$\sum_{j=0}^s \alpha_j \omega^{n+j} = h \sum_{j=0}^s \gamma_j \mathcal{L} \omega^{n+j} + h \sum_{j=0}^{s-1} \beta_j N(\omega^{n+j}), \quad \gamma_s \neq 0. \quad (10)$$

The corrected formulation and the higher-history Adams–Bashforth comparison schemes are instances of (10).

3.2. AB2–CN2. AB2–CN2 is the two-step instance of (10) with $\rho(\zeta) = \zeta^2 - \zeta$, implicit (Crank–Nicolson) weights $\gamma = (0, \frac{1}{2}, \frac{1}{2})$ on $\mathcal{L}$, and explicit (Adams–Bashforth-2) weights $\beta = (-\frac{1}{2}, \frac{3}{2})$ on N [1, 11]. Solving for ω^{n+1} :

$$\omega^{n+1} = r(h) \omega^n + s(h) \left(\frac{3}{2} N^n - \frac{1}{2} N^{n-1} \right), \quad r = \frac{1 + \frac{h}{2} \mathcal{L}}{1 - \frac{h}{2} \mathcal{L}}, \quad s = \frac{h}{1 - \frac{h}{2} \mathcal{L}}. \quad (11)$$

All linear operators are diagonal in Fourier space, so the implicit solve is an elementwise division; the only nonlinearity is the Jacobian, evaluated pseudo-spectrally at five FFTs per call with 2/3 dealiasing. The scheme is second-order accurate in time.

3.3. AB4–CN2. AB4–CN2 keeps the same ρ and implicit CN weights but replaces the explicit polynomial by the four-step Adams–Bashforth one, $\beta = \frac{1}{24}(-9, 37, -59, 55)$:

$$\omega^{n+1} = r(h) \omega^n + s(h) \frac{55N^n - 59N^{n-1} + 37N^{n-2} - 9N^{n-3}}{24}, \quad (12)$$

with the same r, s . Because the CN pair limits the overall order, AB4–CN2 is still second order; what changes is the *composition* of the truncation operators (Sec. 4.1.3) and the explicit stability region (Sec. 4.4).

3.4. Fine-step RK4 reference. The reference trajectory is advanced by classical RK4 at the fine step $h_f = \Delta T / K$. Over one coarse interval, K consecutive RK4 updates define the map $\Phi_{\Delta T}^{\text{RK4}}$. For one fine step,

$$k_1 = G(\omega^n), \quad k_2 = G(\omega^n + \frac{h_f}{2} k_1), \quad k_3 = G(\omega^n + \frac{h_f}{2} k_2), \quad k_4 = G(\omega^n + h_f k_3), \quad (13)$$

$$\omega^{n+1} = \omega^n + \frac{h_f}{6} (k_1 + 2k_2 + 2k_3 + k_4). \quad (14)$$

The accumulated map is used during training and validation; it is not evaluated during corrected coarse-step inference.

4. Analysis of the schemes.

4.1. Error analysis.

4.1.1. The general structure. For an order- p Adams–Bashforth–Crank–Nicolson formula, the order relation (9) implies that the one-step defect against the exact flow is a series starting at h^{p+1} ,

$$\tau_{\text{AB--CN}} = - \sum_{j \geq p+1} \frac{h^j}{D_j} R_j, \quad (15)$$

where the denominators D_j and operators R_j are determined by the scheme's coefficients (α, β, γ) , and each R_j is a fixed linear combination of $\{\mathcal{L}^j \omega, \mathcal{L}^{j-1} N, \mathcal{L}^{j-2} \dot{N}, \dots, N^{(j-2)}\}$: powers of the linear operator acting on the state and the cached nonlinearity, plus time derivatives of the nonlinearity, obtained by matching the scheme's expansion against the exact flow. The analysis below instantiates (15) for AB2–CN2. The corresponding AB4–CN2 expressions are retained only to define a higher-history Adams–Bashforth comparison.

4.1.2. Exact one-step map. Expanding the exact flow in total derivatives (equivalently, the Duhamel integral in φ -functions) and substituting (6),

$$\begin{aligned} \omega(t_n + h) &= \omega + h(\mathcal{L}\omega + N) + \frac{h^2}{2}(\mathcal{L}^2\omega + \mathcal{L}N + \dot{N}) + \frac{h^3}{6}(\mathcal{L}^3\omega + \mathcal{L}^2N + \mathcal{L}\dot{N} + \ddot{N}) \\ &\quad + \frac{h^4}{24}(\mathcal{L}^4\omega + \mathcal{L}^3N + \mathcal{L}^2\dot{N} + \mathcal{L}\ddot{N} + \dddot{N}) \\ &\quad + \frac{h^5}{120}(\mathcal{L}^5\omega + \mathcal{L}^4N + \mathcal{L}^3\dot{N} + \mathcal{L}^2\ddot{N} + \mathcal{L}\dddot{N} + N^{(4)}) + \mathcal{O}(h^6). \end{aligned} \quad (16)$$

4.1.3. The IMEX pair. Taylor-expanding the AB brackets ($\frac{3}{2}N^n - \frac{1}{2}N^{n-1} = N + \frac{h}{2}\dot{N} - \frac{h^2}{4}\ddot{N} + \frac{h^3}{12}\dddot{N} - \dots$), the geometric expansions of r, s , and subtracting from (16):

LTE of AB2–CN2.

$$\tau_{\text{AB2}} = -\frac{h^3}{12}R_3^{(2)} - \frac{h^4}{24}R_4^{(2)} - \frac{h^5}{240}R_5^{(2)} - \frac{h^6}{1440}R_6^{(2)} + \mathcal{O}(h^7), \quad (17)$$

$$R_3^{(2)} = \mathcal{L}^3\omega + \mathcal{L}^2N + \mathcal{L}\dot{N} - 5\ddot{N}, \quad (18)$$

$$R_4^{(2)} = 2\mathcal{L}^4\omega + 2\mathcal{L}^3N + 2\mathcal{L}^2\dot{N} - 4\mathcal{L}\ddot{N} + \dddot{N}, \quad (19)$$

$$R_5^{(2)} = 13\mathcal{L}^5\omega + 13\mathcal{L}^4N + 13\mathcal{L}^3\dot{N} - 17\mathcal{L}^2\ddot{N} + 8\mathcal{L}\dddot{N} - 7N^{(4)}, \quad (20)$$

$$R_6^{(2)} = 43\mathcal{L}^6\omega + 43\mathcal{L}^5N + 43\mathcal{L}^4\dot{N} - 47\mathcal{L}^3\ddot{N} + 28\mathcal{L}^2\dddot{N} - 17\mathcal{L}N^{(4)} + 4N^{(5)}. \quad (21)$$

The orders h^0, h^1, h^2 cancel identically (second-order accuracy).

LTE of AB4–CN2.

$$\tau_{\text{AB4}} = -\frac{h^3}{12}R_3^{(4)} - \frac{h^4}{24}R_4^{(4)} - \frac{h^5}{720}R_5^{(4)} + \mathcal{O}(h^6), \quad (22)$$

$$R_3^{(4)} = \mathcal{L}^3\omega + \mathcal{L}^2N + \mathcal{L}\dot{N}, \quad R_4^{(4)} = 2\mathcal{L}^4\omega + 2\mathcal{L}^3N + 2\mathcal{L}^2\dot{N} + \mathcal{L}\ddot{N}, \quad (23)$$

$$R_5^{(4)} = 39\mathcal{L}^5\omega + 39\mathcal{L}^4N + 39\mathcal{L}^3\dot{N} + 24\mathcal{L}^2\ddot{N} + 9\mathcal{L}\dddot{N} - 251N^{(4)}. \quad (24)$$

Remark 1 (Structural blind spots). $R_3^{(2)} - R_3^{(4)} = -5\ddot{N}$: the AB2 stencil is structurally blind to $\ddot{N}$, and the weight -5 makes $\ddot{N}$ the dominant, irreducible part of the AB2–CN2 closure (quantified in Sec. 4.2). AB4–CN2 zeroes the $\ddot{N}$ slot at orders 3–4 but introduces at h^5 the large $-251 N^{(4)}$ coefficient, the deeper explicit stencil defers, rather than removes, the nonlinear-derivative burden. Both schemes remain second order: the leading $\mathcal{L}$ -monomials are set by the CN pair.

Remark 2 (Padé structure of the linear sector, and a Padé comparison of the full LTE). On the linear sector ($N \equiv 0$) the IMEX pair reduces to pure Crank–Nicolson, whose one-step map $r(h)$ is the diagonal (1,1)-Padé approximant to $e^{h\mathcal{L}}$, $r = \frac{1+h\mathcal{L}/2}{1-h\mathcal{L}/2}$, with truncation series $e^{h\mathcal{L}} - r(h) = -[\frac{(h\mathcal{L})^3}{12} + \frac{(h\mathcal{L})^4}{12} + \frac{13(h\mathcal{L})^5}{240} + \frac{43(h\mathcal{L})^6}{1440}] + \dots$ (verified symbolically); the $\mathcal{L}^p\omega$ coefficients of (17), $\{-\frac{1}{12}, -\frac{1}{12}, -\frac{13}{240}, -\frac{43}{1440}\}$, are exactly these Padé error coefficients. This identifies *which part* of the LTE a higher Padé scheme could remove: replacing CN by the (2,2)-Padé approximant $R_{22}(z) = (1 + \frac{z}{2} + \frac{z^2}{12}) / (1 - \frac{z}{2} + \frac{z^2}{12})$, whose error is $e^z - R_{22}(z) = \frac{z^5}{720} + \mathcal{O}(z^6)$, would push the entire $\mathcal{L}^p\omega$ column of the truncation table to $\mathcal{O}(h^5)$, but leaves every N -derivative column of (17)–(22) (in particular the dominant $-5\ddot{N}$ slot) untouched, since those originate in the explicit AB bracket, not in the linear propagator. Why does Padé address only the linear part? A Padé approximant is a rational function of a *fixed operator*, and reproduces its exponential, the flow of the linear, autonomous sector. The nonlinear flow is generated by no fixed operator: the N -derivative columns are trajectory-dependent, so no rational function of $\mathcal{L}$ (or of any frozen operator) can represent them. The discrete-time analogues on the nonlinear side are exponential and ETD constructions [8, 20], which reduce but do not close the nonlinear-derivative error. A Padé upgrade is therefore orthogonal to the closure constructed here: it removes the analytically available linear column, while the closure supplies the computationally intensive nonlinear-derivative columns. All coefficients in (17)–(22) were verified symbolically.

4.1.4. Numerical verification. For a developed-flow snapshot ω^n , we form a converged reference history and future, take one AB2–CN2 step of size h , and measure the empirical defect and the residuals after subtracting the analytical prediction kept through order P . Keeping $R_3, \dots, R_P$ leaves a residual $\mathcal{O}(h^{P+1})$; the measured log–log slopes are 4, 5, 6 for $P = 3, 4, 5$, confirming R_3, R_4, R_5 (R_6 ’s slope-7 line sits at the float64 roundoff floor and is confirmed symbolically instead). The chain-rule derivatives in (5) are independently validated against converged temporal finite differences. Figure 1 records the corresponding asymptotic orders after normalization at a common reference step.

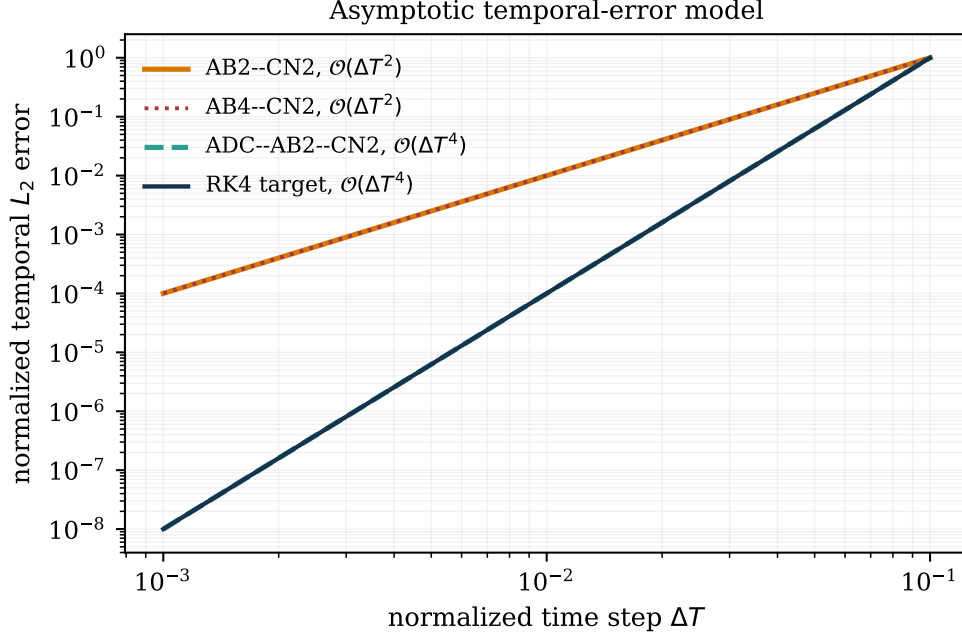


Figure 1: Normalized asymptotic temporal L_2 -error model. AB2–CN2 and AB4–CN2 retain second-order behavior because of the Crank–Nicolson pairing, whereas the order-four defect correction follows the RK4 target through the assembled order. The curves display theoretical slopes and do not encode case-specific error constants.

4.2. Magnitudes, the convergence radius, and error propagation. Consider the LTE of an order- p Adams–Bashforth–Crank–Nicolson formula with respect to the exact flow,

$$\tau_{\text{AB--CN}} = - \sum_{j \geq p+1} \frac{h^j}{D_j} R_j. \quad (25)$$

Everything in this subsection applies to the Adams–Bashforth family. The measurements use AB2–CN2, whose (R_j, D_j) are those of (17). The usability of this expansion depends on three questions.

- (a) The growth of the N -derivatives determines the convergence radius of the series. Section 4.2.2 relates this radius to the Taylor microscale of the nonlinear term.
- (b) The admissible history depth is constrained by temporal decorrelation. States cease to inform the derivative estimate once the stencil span exceeds the local correlation horizon (Sec. 4.2.3).
- (c) The derivative-estimation error propagates through the assembled correction according to the coefficient structure derived in Sec. 4.2.4.

We develop the theory of each question in turn; all measurements are collected in Sec. 4.2.5.

4.2.1. Finite-radius condition. The defect of one Adams–Bashforth–Crank–Nicolson step is a power series in the step,

$$\delta(\Delta T) = \sum_{j \geq p+1} c_j \Delta T^j, \quad c_j = \frac{R_j}{D_j} \quad (\text{for AB2--CN2: } D_j = \{12, 24, 240, 1440\}, j = 3, \dots, 6). \quad (26)$$

Its radius of convergence is given by Cauchy–Hadamard,

$$\frac{1}{\Delta T^\star} = \limsup_{j \rightarrow \infty} \|c_j\|^{1/j}. \quad (27)$$

The classical root/ratio comparison [27, Ch. 3] supplies the analytic bounds used to bracket the measurement: for any positive sequence,

$$\liminf_j \frac{\|c_{j+1}\|}{\|c_j\|} \leq \liminf_j \|c_j\|^{1/j} \leq \limsup_j \|c_j\|^{1/j} \leq \limsup_j \frac{\|c_{j+1}\|}{\|c_j\|}, \quad (28)$$

so the reciprocal successive-ratio interval *brackets* ΔT^* while the root test estimates it.

Proposition 1 (Finite radius). *ΔT^* is finite and set by the growth of the N -derivatives. For $\Delta T > \Delta T^*$ no finite-order analytic closure converges: past ΔT^* , adding correction orders makes the error worse, regardless of how many orders are carried.*

This overturns the naive expectation that higher-order corrections extend the usable step. Measured radii and brackets per member are in Table 2 (Sec. 4.2.5).

4.2.2. The Taylor microscale of the nonlinear term. View $N(t)$ as a stationary process with the domain inner product and define the autocorrelation $\rho(\tau) = \mathbb{E}\langle N(t), N(t+\tau) \rangle / \mathbb{E}\|N\|^2$. Expanding $N(t+\tau) = N + \tau\dot{N} + \frac{\tau^2}{2}\ddot{N} + \mathcal{O}(\tau^3)$ inside the inner product,

$$\mathbb{E}\langle N, N(t+\tau) \rangle = \mathbb{E}\|N\|^2 + \tau \mathbb{E}\langle N, \dot{N} \rangle + \frac{\tau^2}{2} \mathbb{E}\langle N, \ddot{N} \rangle + \mathcal{O}(\tau^3). \quad (29)$$

Stationarity kills the linear term, $\mathbb{E}\langle N, \dot{N} \rangle = \frac{1}{2} \frac{d}{dt} \mathbb{E}\|N\|^2 = 0$; and differentiating that same identity once more in t , $0 = \frac{d}{dt} \mathbb{E}\langle N, \dot{N} \rangle = \mathbb{E}\|\dot{N}\|^2 + \mathbb{E}\langle N, \ddot{N} \rangle$, i.e. $\mathbb{E}\langle N, \ddot{N} \rangle = -\mathbb{E}\|\dot{N}\|^2$. Hence the exact identity (no Gaussianity, no spectral assumption):

$$\boxed{\rho(\tau) = 1 - \frac{\tau^2}{2} \sigma^2 + \mathcal{O}(\tau^4), \quad \sigma^2 = \frac{\mathbb{E}\|\dot{N}\|^2}{\mathbb{E}\|N\|^2}, \quad \tau_\lambda = \frac{1}{\sigma}.} \quad (30)$$

τ_λ is the Taylor microscale of the N -process: the osculating-parabola (curvature) definition of the microscale from the correlation function goes back to Taylor [32] for spatial velocity correlations and is standard in the turbulence literature [25, 34]; here it is applied to the *temporal* autocorrelation of the nonlinear term. The same identity holds for ω and per wavenumber shell, with $\sigma(\kappa) = \|\dot{\omega}_\kappa\| / \|\omega_\kappa\|$. Per-order growth ladders as the moment ratio

$$\sigma_m^2 = \frac{\int \sigma(\kappa)^{2(m+1)} E_N(\kappa) d\kappa}{\int \sigma(\kappa)^{2m} E_N(\kappa) d\kappa}, \quad (31)$$

which increases with m because $\sigma(\kappa)$ increases with κ : higher derivatives progressively sample the fast small-scale tail.

The relationship between the radius and the microscale follows from the same expansion. The truncation series (26) is the Taylor expansion of the one-step defect in the step size; its terms are controlled by the Taylor expansion of $N(t)$ itself, which (30) shows is governed by the single time scale $\tau_\lambda = 1/\sigma$: the parabola $1 - \tau^2 \sigma^2 / 2$ reaches zero at $\tau = \sqrt{2} \tau_\lambda$, i.e. the nonlinear term decorrelates from itself, and its Taylor polynomial ceases to represent it, once the lag is $\mathcal{O}(\tau_\lambda)$. With $\|N^{(m)}\| \sim \sigma_m^m \|N\|$ and the denominators D_p contributing $D_p^{1/p} = \mathcal{O}(1)$ growth over the measured orders, Cauchy–Hadamard (27) then gives:

Proposition 2 (Radius–microscale law). *$\Delta T^* = C \tau_\lambda$ with C an $\mathcal{O}(1)$ constant set by the competition between the denominator growth $D_p^{1/p}$ and the ladder escalation (31). All regime dependence of ΔT^* enters through the single measurable rate σ : the analytic closure’s convergence horizon is a fixed small multiple of the Taylor microscale of the nonlinear term.*

The measured constants per member, and the verification of (30) itself, are in Sec. 4.2.5.

4.2.3. Finite-history condition. The closure estimates N -derivatives from S stored marks at lags $\{0, \Delta T, \dots, (S-1)\Delta T\}$. Depth raises the approximation order ($\mathcal{O}(\Delta T^{S-m})$ for the m -th derivative) but reaches further back in time; old marks are stale once the span approaches ΔT^* .

Deeper stencils lower the N -derivative error while the stencil span stays well inside ΔT^* , and raise it once the span exceeds ΔT^* ; the usable step of a depth- S closure scales as $\Delta T_{\text{inner}} \sim \Delta T^*/(S-1)$. Consequence for pooled training: a regime-blind pool caps at the most restrictive member’s correlation horizon; tiers beyond that horizon are retained to regularize the estimator under adverse conditions.

4.2.4. Error propagation. An error ε on the learned field enters the assembled closure through the amplification expression

$$\Delta\delta_{p,k} = \frac{\Delta T^p}{D_p} |a_{p,k}| \varepsilon \left\| \mathcal{L}^{p-k} \text{field}_k \right\|, \quad (32)$$

so terms carrying powers of $\mathcal{L}$ are potentially amplified. Whether they are, in practice, is a question about the measured sizes of the fields: Table 3 (Sec. 4.2.5) shows that across the entire trained envelope $\Delta T \leq 1.5 \times 10^{-2}$ the closure error equals the relative $\ddot{N}$ error, one-to-one and un-amplified. Any improvement in $\ddot{N}$ accuracy converts to closure-error reduction 1:1, and $\ddot{N}$ accuracy is the single quantity that sets the rollout ceiling. Two design consequences: the equal-weight per-order relative- L_2 loss is the right objective across the envelope (no amplification weighting needed), and the $\ddot{N}$ validation error is therefore the principal diagnostic for the rollout ceiling.

4.2.5. Measurements. All solver-side measurements use Float64 arithmetic and 2/3-dealiased Jacobians matching the solver right-hand side, 512² forced-turbulence members at marks $h_f = 5 \times 10^{-3}$; the reference is the analytic spectral chain rule (5); errors are relative L_2 in the resolved band. Every member integrates (1) with the static deterministic forcing

$$F(x, y) = A[\cos(k_f x) + \cos(k_f y)], \quad (33)$$

and differs only in (F, k_f, ν, β) :

member	F	k_f	ν	β
FRC-Re25k	$A[\cos(k_f x) + \cos(k_f y)]$	–	4.0×10^{-5}	1.0
FRC-combo	$A[\cos(k_f x) + \cos(k_f y)]$	–	4.0×10^{-5}	0.5
FRC-kf4	$A[\cos(k_f x) + \cos(k_f y)]$	4	1.02×10^{-4}	1.0

Radius (Sec. 4.2.1). The root test estimates ΔT^* (root-test median over $p = 3, \dots, 6, 32$ anchors per member) and the ratio interval (28) brackets it: The per-order growth ratios *increase* with m

Table 2: Measured convergence radii per member.

member	$\ N^{(m+1)}\ / \ N^{(m)}\ , m = 0, \dots, 4$	ΔT^* (root test)	ratio bracket
Re25k	34, 58, 72, 84, 93	0.066	[0.017, 0.136]
combo	15, 27, 37, 45, 51	0.139	[0.033, 0.270]
kf4	10, 19, 28, 33, 37	0.199	[0.043, 0.366]

(34 $\rightarrow$ 93 for Re25k): the ladder escalation (31) in action.

Microscale (Sec. 4.2.2). The identity (30) holds to 15% on N and 3% on ω (curvature fit vs $1/\sigma$; Re25k $\tau_\lambda \sigma_N = 1.15$, kf4 1.09; on ω , 0.98 and 0.97). Two further measured facts: the state decorrelates an order of magnitude more slowly than its nonlinear term ($\sigma_N/\sigma_\omega \approx 7.5$ –11; over 27 lags $\rho_\omega \geq 0.96$ while ρ_N falls to 0.2–0.6), so a time stencil on the (ω, ψ) -history sees smooth data; and ρ_N plateaus at a positive value (≈ 0.2 , the static forcing plus time-mean of $-\mathcal{J}$), which biases integral decorrelation times but not τ_λ . The measured radius–microscale constants (Prop. 2):

	Re25k	combo	kf4
$C = \Delta T^*/\tau_\lambda$	2.27	2.05	1.92

On these data $C = 2.08 \pm 0.15$, constant to 7% across $2.5\times$ in Re and $2\times$ in β : the convergence horizon is two Taylor microscales of the nonlinear term.

Error propagation (Sec. 4.2.4). Measured shares of the assembled correction error $\|\delta\|$ carried by each slot of (32) (at $\Delta T = 5 \times 10^{-3}$ the per-order errors are 2.6/3.0/4.0% on $\dot{N}/\ddot{N}/\ddot{N}$ and the total error is 3.00%, exactly the $\ddot{N}$ error; the 1.5×10^{-2} row uses unit per-term errors, i.e. pure amplification geometry):

Table 3: Amplification shares.

ΔT	member(s)	$\mathcal{L}^0 \ddot{N}$	$\mathcal{L}^0 \ddot{N}$	$\mathcal{L}^1 \dot{N}$	$\mathcal{L}^2 \dot{N}$
5×10^{-3}	all three	$\approx 100\%$	residual, $< 1\%$ combined		
1.5×10^{-2}	Re25k	100.2%	2.6%	0.5%	0.00%

4.3. Cost analysis. The general accounting is set by the Adams–Bashforth history depth. An s -step explicit Adams–Bashforth update pays one new nonlinear evaluation per step and stores s cached history levels. Adams–Bashforth–Moulton predictor–corrector pairs instead pay two nonlinear evaluations [13]. In a pseudo-spectral code the nonlinear evaluation is a fixed number of FFTs, here five per Jacobian (two inverse transforms of $\nabla\psi$, two of $\nabla\omega$, one forward of the dealiased product; all other operations are Fourier-diagonal multiplies), and the implicit solve of the IMEX pairing is an elementwise division. For the representative schemes (storage counts persistent per-step state, fields of size $N_x \times N_y$):

scheme	run time per step	storage
AB2–CN2	≈ 5 FFTs	$\omega^n + 2$ cached N levels
AB4–CN2	≈ 5 FFTs	$\omega^n + 4$ cached N levels
ADC–AB2–CN2	≈ 5 FFTs + one shallow network	$\omega^n + S$ cached levels

Cost comparisons at matched accuracy include transfers between the Float64 solver and the Float32 estimator.

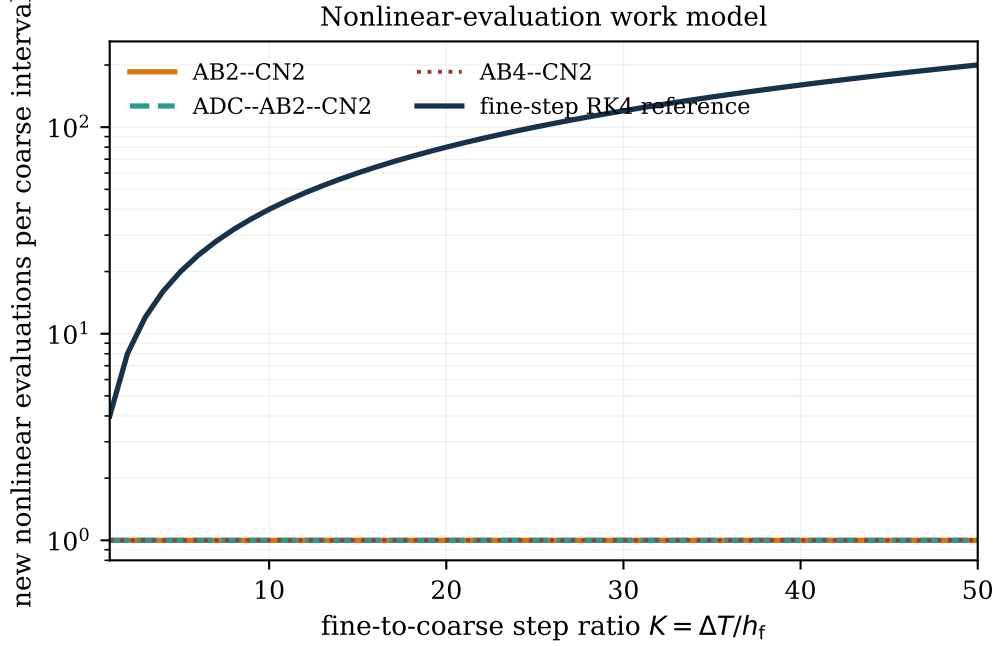


Figure 2: Nonlinear-evaluation work over one coarse interval. The Adams–Bashforth–Crank–Nicolson updates require one new nonlinear evaluation, whereas an RK4 reference advanced with K fine steps requires $4K$ evaluations. The Float32 estimator adds one shallow-network call but no additional pseudo-spectral Jacobian evaluation.

4.4. Stability analysis. The general tools apply to the Adams–Bashforth schemes of Sec. 3.1. Their absolute-stability region is $\{z : |g(z)| \leq 1\}$ with $g(z)$ the max-modulus root of $\rho(\zeta) - z\sigma(\zeta)$; for a one-step scheme, g is its stability function. For any scheme, the truncation series and the amplification factor are two views of the same object: the truncation *symbol* is the amplification gap $\hat{\tau}(z) = e^z - g(z)$. We now evaluate these for the Adams–Bashforth comparison family.

4.4.1. Explicit regions. For an explicit Adams family, the absolute-stability region is $\{z : |g(z)| \leq 1\}$, where g is the maximum-modulus root of $\rho(\zeta) - z\sigma(\zeta)$. Advective eigenvalues are nearly imaginary, so the imaginary-axis stability boundary is particularly informative [13]. AB2 is marginal at the origin on the imaginary axis, whereas higher-history Adams formulas introduce finite intervals whose size depends nonmonotonically on order. In the IMEX pairing, Crank–Nicolson treatment of viscosity displaces the coupled modal spectrum from the imaginary axis.

Analytical absolute-stability boundaries

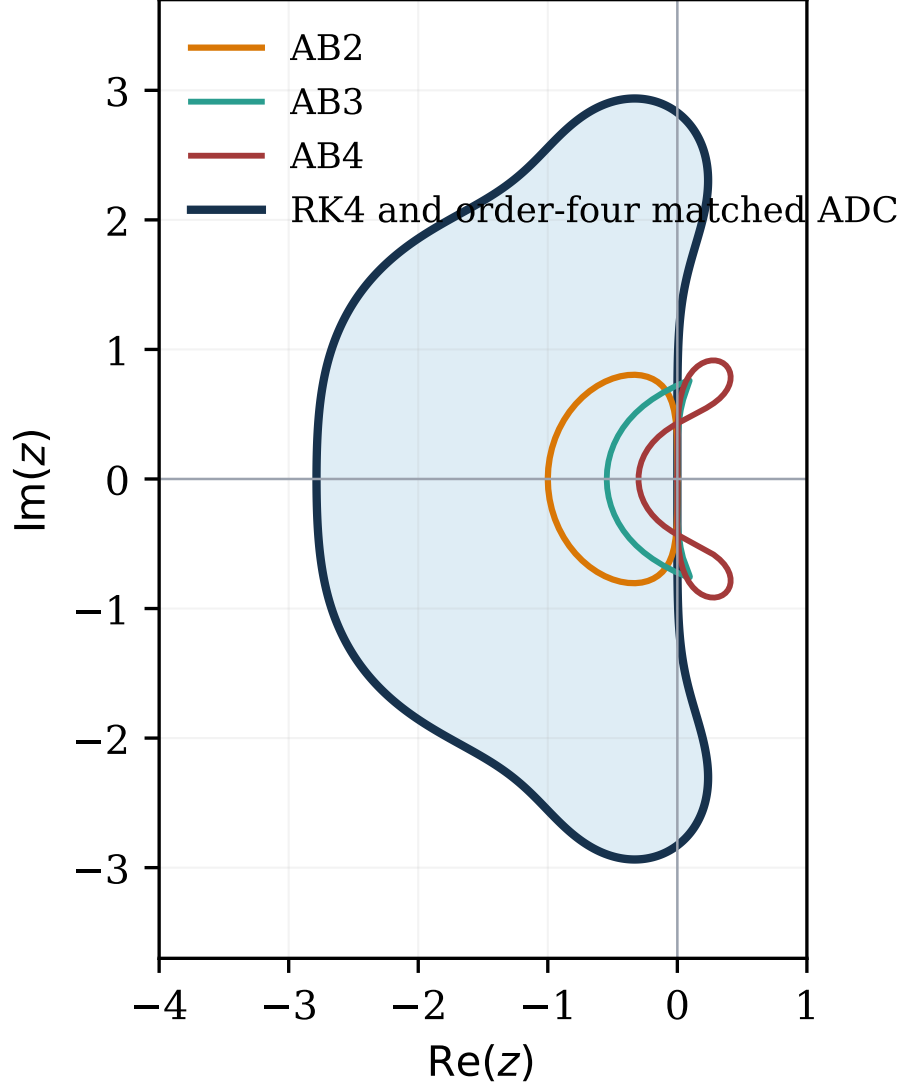


Figure 3: Analytical absolute-stability boundaries for the explicit Adams–Bashforth formulas and RK4. Through the assembled order, the corrected AB2–CN2 amplification factor follows the RK4 target; the shaded region therefore indicates the target stability geometry inherited by the finite-order correction.

4.4.2. IMEX amplification factors. On a Fourier mode, split $z = z_L + z_N$ with $z_L = h\hat{\mathcal{L}}(\mathbf{k})$ the implicitly treated linear symbol and z_N the explicit advection symbol; the frozen-coefficient (von Neumann) worst case over background velocity is $z_N = ihU|\mathbf{k}|$. For an Adams–Bashforth–Crank–Nicolson scheme (10), the normal-mode ansatz $\omega^{n+j} = G^j \omega^n$ gives the degree- s amplification polynomial

$$\rho(G) - z_L \gamma(G) - z_N \sigma(G) = 0, \quad (34)$$

with ρ, γ, σ the generating polynomials of Sec. 3.1 and g its max-modulus root. For the CN-implicit pairing shared by the comparison family, dividing (34) by its leading coefficient $1 - z_L/2$ reduces

everything to two scalars,

$$r = \frac{1 + z_L/2}{1 - z_L/2}, \quad \tilde{\rho} = \frac{z_N}{1 - z_L/2} \quad (35)$$

(note r is Crank–Nicolson-specific, it is the $(1, 1)$ -Padé map of the implicit weights $\gamma = (0, \frac{1}{2}, \frac{1}{2})$, cf. Remark 2, while a different implicit pairing changes r accordingly; $\tilde{\rho}$ is written with a tilde to avoid collision with the generating polynomial $\rho(\zeta)$). AB2–CN2 is then the two-step recurrence $\omega^{n+1} = (r + \frac{3}{2}\tilde{\rho})\omega^n - \frac{1}{2}\tilde{\rho}\omega^{n-1}$ with companion amplification

$$g^2 - (r + \frac{3}{2}\tilde{\rho})g + \frac{1}{2}\tilde{\rho} = 0, \quad g_{\text{AB2}} = \text{max-modulus root}, \quad (36)$$

AB4–CN2 has the analogous quartic companion with the same $r, \tilde{\rho}$ and the AB4 bracket weights $\beta = \frac{1}{24}(-9, 37, -59, 55)$. Absolute stability of the discrete mode is $|g| \leq 1$; the maximum stable ΔT per regime follows by bisection over all dealiased modes, using each regime’s measured characteristic velocity.

4.4.3. Truncation symbols. For the closure analysis we need the LTE in amplification space. Replacing $\mathcal{L}^j \omega \rightarrow \lambda_L^j$ and $N^{(m)} \rightarrow \lambda_N(\lambda_L + \lambda_N)^m$ per mode, the order- p operator of any scheme in the family becomes the polynomial

$$P_p(z_L, z_N) = a_{p,0}z_L^p + \sum_{k=1}^p a_{p,k}z_L^{p-k}z_N(z_L + z_N)^{k-1}. \quad (37)$$

Here $a_{p,k}$ is the coefficient of the k -th slot of the order- p operator R_p , the entries of its coefficient row, read left to right, and a superscript tags the scheme: $a_{p,k}^{(2)}$ are the rows of (17) (e.g. the order-3 row of AB2–CN2 is $a_{3,\cdot}^{(2)} = (1, 1, 1, -5)$), the coefficients of $\mathcal{L}^3\omega, \mathcal{L}^2N, \mathcal{L}\dot{N}, \ddot{N}$ in $R_3^{(2)}$, and $a_{p,k}^{(4)}$ those of (22). With the AB2–CN2 rows,

$$\hat{\tau}_{\text{AB2}} = e^z - g_{\text{AB2}} = - \sum_p \frac{P_p}{D_p}. \quad (38)$$

5. Amortized defect correction and full numerical scheme.

5.1. RK4-matched AB2–CN2 defect correction. Let $\Phi_{\Delta T}$ denote the exact flow map, let $\Psi_{\Delta T}^{\text{AB2CN2}}$ denote one coarse AB2–CN2 update, and let $\Phi_{\Delta T}^{\text{RK4}} = (\Psi_{h_f}^{\text{RK4}})^K$ denote the RK4 reference accumulated with $h_f = \Delta T/K$. The two local errors satisfy

$$\tau_{\text{AB2}} = \Phi_{\Delta T} - \Psi_{\Delta T}^{\text{AB2CN2}}, \quad \tau_{\text{RK4}}^{(K)} = \Phi_{\Delta T} - \Phi_{\Delta T}^{\text{RK4}}. \quad (39)$$

Their difference is therefore the correction that maps the coarse Adams–Bashforth trajectory to the accumulated RK4 trajectory,

$$\delta_{\text{RK4}}^{(K)} = \Phi_{\Delta T}^{\text{RK4}} - \Psi_{\Delta T}^{\text{AB2CN2}} = \tau_{\text{AB2}} - \tau_{\text{RK4}}^{(K)}. \quad (40)$$

The AB2–CN2 contribution is

$$\tau_{\text{AB2}} = -\frac{h^3}{12}R_3 - \frac{h^4}{24}R_4 - \frac{h^5}{240}R_5 - \frac{h^6}{1440}R_6 + \mathcal{O}(h^7), \quad (41)$$

where the operators R_j are given in Sec. 4.1.3. The RK4 contribution begins at fifth order for $K = 1$ and decreases as the fine step is refined. Consequently, the order-three and order-four correction

operators are determined entirely by the AB2–CN2 defect, while the target-specific term enters at the first RK4 defect order. Truncation after order P gives

$$\omega_{\text{ADC}}^{n+1} = \Psi_{\Delta T}^{\text{AB2CN2}}(\omega^n, \omega^{n-1}) - \sum_{p=3}^P \frac{\Delta T^p}{D_p} \hat{R}_p - \hat{\tau}_{\text{RK4}}^{(K)}, \quad 3 \leq P \leq 6. \quad (42)$$

Here $\hat{\tau}_{\text{RK4}}^{(K)}$ contains only the target terms retained at the selected order. The estimator $\hat{R}_p$ replaces the nonlinear temporal-derivative slots, whereas all powers of $\mathcal{L}$ and all contributions involving the available state and nonlinear history are evaluated analytically. Thus the correction is defined by an explicit numerical target rather than by a direct regression to the exact flow.

5.2. Analytical and learned contributions. Each truncation operator has the decomposition

$$R_p = \underbrace{R_p^{\text{anal}}}_{\text{Fourier-diagonal operations}} + \underbrace{R_p^{\text{nl}}}_{\text{temporal derivatives of } N}. \quad (43)$$

In Fourier space, $\mathcal{L}^j$ is a pointwise multiplication by the precomputed symbol $\hat{\mathcal{L}}(\mathbf{k})^j$. The analytical contribution therefore requires no additional Jacobian evaluation. The network supplies $\hat{R}_p^{\text{nl}}$ from the stored history, and the complete estimator is $\hat{R}_p = R_p^{\text{anal}} + \hat{R}_p^{\text{nl}}$.

5.3. Full numerical scheme. At each coarse step, the algorithm evaluates $N(\omega^n)$ in Float64, advances the AB2–CN2 backbone in Float64, casts the normalized history to Float32 for the shallow network, casts the predicted derivative channels back to Float64, and assembles the defect correction in Float64. Thus

$$\omega_{\text{base}}^{n+1} = r(\Delta T)\omega^n + s(\Delta T)\left(\frac{3}{2}N^n - \frac{1}{2}N^{n-1}\right), \quad (44)$$

$$\omega_{\text{ADC}}^{n+1} = \omega_{\text{base}}^{n+1} - \sum_{p=3}^P \frac{\Delta T^p}{D_p} \left(R_p^{\text{anal}} + \hat{R}_{p,\text{FP32}}^{\text{nl}}\right) - \hat{\tau}_{\text{RK4}}^{(K)}. \quad (45)$$

The lower-precision network never updates the prognostic state directly; its output enters only after conversion to Float64 and multiplication by the corresponding defect coefficient.

5.4. Inference decomposition. The per-step work consists of one pseudo-spectral nonlinear evaluation, one diagonal Crank–Nicolson solve, inexpensive diagonal defect operations, and one shallow Float32 network evaluation. The stored nonlinear history is reused by both AB2–CN2 and the derivative reconstruction. Consequently, the correction does not introduce a second nonlinear Jacobian evaluation.

6. S -stage bilinear convolution approximation of higher-order derivatives.

6.1. Network architecture. The estimator maps the stored step history to the local derivative fields,

$$X = [\omega_0, \dots, \omega_{-(S-1)}, \psi_0, \dots, \psi_{-(S-1)}] \mapsto [\dot{N}, \ddot{N}, \ddot{N}], \quad (46)$$

in three stages that mirror the operator being estimated. Throughout, S denotes the (user-defined) time-stencil depth, W the (user-defined, odd) spatial-stencil width, and p the highest assembled derivative order.

1. **Time finite differences (frozen, exact).** Backward-FD rows V_{ki} from the Vandermonde system $\sum_i V_{ki} \frac{(-i)^p}{p!} = \delta_{kp}$ produce $\omega_{\text{FD}}^{(k)}, \psi_{\text{FD}}^{(k)}$, $k = 0, \dots, p$, via the per-sample path $V_{ki}/\Delta T^k$. Nothing is learned here: rescaling the unit stencil per sample is exact; each emitted row is fed by all S snapshots, and the order- k row of a depth- S stencil carries accuracy $\mathcal{O}(\Delta T^{S-k})$.
2. **Learnable spatial gradients.** Depthwise width- W stencils (central-difference initialization of order $W-1$) apply ∂_x, ∂_y to the $2(p+1)$ differentiated fields, circular padding. The stencil parameter is the *dimensionless* unit-spacing operator; the physical $1/dx, 1/dy$ scaling is applied per sample at forward, exact by linearity, and the optimizer never sees the $1/dx$ amplification. One stencil pair serves all grids.
3. **Jacobian features and physics-initialized mix.** The $(p+1)^2$ pointwise products $\mathcal{J}(\psi^{(i)}, \omega^{(j)})$ feed a 1×1 mix initialized to the exact chain-rule binomials of (5) ($w[m-1, (m-j, j)] = -\binom{m}{j}$): at initialization the network *is* the analytic estimator up to the two FD truncations.

The parameter count is a closed function of the structural sizes: the depthwise gradient stencils carry $2(p+1) \cdot 2 \cdot W^2$ weights (two fields at $p+1$ orders, two directions, $W \times W$ kernels), the 1×1 mix $(p+1)^2 \cdot p$, and the frozen time-FD buffer $S(p+1)$ inert weights (not trained capacity); the trained capacity is the stencils and the mix. There are no FFTs inside the model, the pointwise product is the sole irreducible nonlinearity, and carrying ψ in the history makes even the β term local ($\beta \partial_x \Delta^{-1} \omega = -\beta v$). The architecture is exactly quadratic in the input field, matching the quadratic nonlinearity of (3).

Remark 3 (Why the spatial stencils are learnable at all). They serve two roles: they approximate ∂_x, ∂_y and absorb the differentiated temporal finite-difference error. The second role produces the Wiener mechanism analyzed next and explains why the width W matters and why the stencils are conditioned in Sec. 6.3.

6.2. Exact error analysis of the estimator. Two error operators separate cleanly. The time-FD estimate carries

$$\varepsilon_k \equiv \omega_{\text{FD}}^{(k)} - \omega^{(k)} = C_k \Delta T^{S-k} \omega^{(S)} + \mathcal{O}(\Delta T^{S-k+1}), \quad C_k = \frac{1}{S!} \sum_{i=0}^{S-1} V_{ki} (-i)^S, \quad (47)$$

with $\varepsilon_0 = 0$ and $\varepsilon_k^\psi = \Delta^{-1} \varepsilon_k$ (Δ^{-1} commutes with the stencil). The learned gradients carry $D_\theta = \partial + \delta_\theta$ with δ_θ the stencil gap. Propagating both through the bilinear Jacobian and discarding $\mathcal{O}(\varepsilon \delta, \varepsilon^2, \delta^2)$,

$$\Delta N^{(m)} = T^{(m)} + \mathcal{A}^{(m)}[\delta_\theta], \quad T^{(m)} = \Delta T^{S-m} C_m \left[-\mathcal{J}(\Delta^{-1} \omega^{(S)}, \omega) - \mathcal{J}(\psi, \omega^{(S)}) \right] + \mathcal{O}(\Delta T^{S-m+1}) : \quad (48)$$

the leading time error *at every order* m is driven by the same field $\omega^{(S)}$, entering through exact Jacobians, while $\mathcal{A}^{(m)}[\delta_\theta]$ is linear in the stencil gap. Training therefore solves a Wiener *filtering* problem, the least-squares-optimal *linear* correction of one signal from another [21, 38]: the optimal stencil correction δ^* is the Wiener filter that pre-cancels the time error it can reach.

6.2.1. The pooled floor and its three parts. We first state the training objective. The pool consists of tiers $j = (\text{member}, \Delta T)$ with weights w_j . Training minimizes the quadratic loss $\mathcal{L}(\delta) = \sum_j w_j \mathbb{E}_j \|\Delta N^{(m)}(\delta)\|^2$ over a single, shared width- W stencil correction δ . The question is: how small can this loss get? Orthogonal decomposition about the per-tier optima gives the answer in

three parts:

$$\mathcal{L}_{\min}^{\text{uncond}} = \underbrace{\sum_j w_j \mathcal{L}_j(\delta_j^{\star\star})}_{\text{(i) cascade remainder}} + \underbrace{\min_{\bar{\delta}} \sum_j w_j \mathbb{E}_j \|\mathcal{A}[\delta_j^{\star} - \bar{\delta}]\|^2}_{\text{(ii) pooled-variance mismatch}} + \underbrace{\sum_j w_j \mathbb{E}_j \|\mathcal{A}[\delta_j^{\star\star} - \delta_j^{\star}]\|^2}_{\text{(iii) width-}W \text{ shape deficit}}, \quad (49)$$

Here $\delta_j^{\star}$ is the best width- W filter for tier j , and $\delta_j^{\star\star}$ is the best shell-diagonal transfer. The three parts have distinct interpretations. Part (i) is the cascade remainder: error no filter of any width can reach. Part (ii) is the pooled-variance mismatch: the best filter differs from tier to tier, and one shared stencil can only fit the pooled mean. Part (iii) is the shape deficit: what a width- W filter misses even on a single tier. Conditioning removes the pooled-variance component by allowing the local filter to depend on the measurable decorrelation rate. Equation (49) therefore distinguishes the irreducible cascade remainder from the approximation error associated with a finite convolution width.

6.2.2. Single-mode analysis of the driver. Equation (48) says the time error at every order is driven by the single field $\omega^{(S)}$. For the stencil to absorb that error, we need to know how $\omega^{(S)}$ is related, shell by shell, to the resolved fields the stencil actually sees. This subsection derives that relation under four labeled assumptions, each with classical support:

- (A1) *Sweeping split*, $\hat{N} = -i(\mathbf{k} \cdot \mathbf{U})\hat{\omega} + \hat{f}$: small scales are advected (“swept”) past a fixed point by the energetic large-scale velocity $\mathbf{U}$, and this sweeping dominates the Eulerian decorrelation of high- κ modes [7, 23, 35]. The cascade input $\hat{f}$ collects everything else.
- (A2) *Frozen coefficients over the stencil span*: $\mathbf{U}$ and $\Omega = \mathbf{k} \cdot \mathbf{U}$ are treated as constant over the $(S-1)\Delta T$ span, Taylor’s frozen-turbulence hypothesis applied on a short window [33].
- (A3) *Advective band*: $\Omega^2 \gg \lambda^2$, i.e. the sweeping frequency dominates the viscous/drag rate on the shells that matter.
- (A4) *Subdominant triads* in the shell balance.

Under (A1)–(A4), induction on $\hat{\omega}^{(n)} = a^n \hat{\omega} + \sum_j a^j \hat{f}^{(n-1-j)}$, $a = i\Omega + \lambda$, gives the per-shell transfer between derivative orders:

$$\widehat{\omega^{(S)}}_{\kappa} = \sigma_{\omega}(\kappa)^{S-k} e^{i\phi(\kappa)} \widehat{\omega^{(k)}}_{\kappa} + \hat{r}, \quad \phi = (S-k)\frac{\pi}{2} \text{sign}(\Omega), \quad \hat{r} = \hat{r}_1 + \hat{r}_2 + \hat{r}_3. \quad (50)$$

Equation (50) is the discrete statement of sweeping decorrelation [7, 23]: each time derivative multiplies a swept mode by its sweeping frequency and rotates its phase by $\pi/2$. The shell rate is

$$\sigma_{\omega}(\kappa)^2 = \langle \Omega^2 \rangle_{\text{shell}} = \frac{1}{2} \kappa^2 U^2 + \frac{1}{2} \beta^2 / \kappa^2 - U \beta \cos \alpha_{\mathbf{U}}, \quad (51)$$

whose three terms have distinct classical origins: the first is the shell average of the squared sweeping frequency $(\mathbf{k} \cdot \mathbf{U})^2$ [7]; the second is the shell average of the squared Rossby-wave frequency $\beta \mathbf{k}_x / \kappa^2$ of the β -plane term [26]; the third is their cross-correlation, controlled by the angle $\alpha_{\mathbf{U}}$ between the sweep and the zonal direction.

The remainder taxonomy bounds what any conditioning can recover:

- r_1 : the cascade input, shell-incoherent with $\omega^{(k)}$. *Irreducible* by any stencil (measured incoherent fraction 0.71–0.79 on well-resolved tiers).
- r_2 : coefficient drift within the stencil span, the failure of (A2), of size $\frac{(S-k)(S+k-1)}{2} \tau_{\kappa} / T_U$, with $\tau_{\kappa} = 1/\sigma_{\omega}(\kappa)$ the per-shell state timescale and T_U the large-eddy turnover. Conditionable in time.
- r_3 : intra-shell anisotropy. For odd $S-k$ the sweeping transfer $\propto \cos^{S-k} \theta$ shell-averages to *zero*, so an isotropic correction captures none of it; only $\cos 3\theta, \cos 5\theta$ harmonics are reachable by angular features.

6.3. Physics conditioning. Combining (47) and (50), the per-shell regression coefficient of the time error on the resolved field factorizes:

$$r_j^{(k)}(\kappa) = \underbrace{\Delta T_j^{S-k}}_{\text{exact}} \times \underbrace{C_k \sigma_\omega(\kappa; \text{Re}, \beta, \mu)^{S-k} e^{i\phi(\kappa)}}_{g^{(k)}(\kappa)} + \text{h.o.t.} \quad (52)$$

Two structural facts follow. First, the optimal stencil correction varies as $(\Delta T \sigma(\kappa))^{S-k}$, a factor $3^5 \approx 243$ across our ΔT -sweep at $k = 2$, which *derives* the physics conditioning: no single unconditioned stencil can track it. Second, (Re, β, μ) enter only through $\sigma_\omega(\kappa)$, so the regime never needs to be supplied as scalars: it is measurable from the model’s own input stack. The two-mark estimator with arcsin de-biasing (a single mode rotating at Ω has FD response $\frac{2}{\Delta T} |\sin(\Omega \Delta T/2)|$),

$$\hat{\sigma}(\kappa) = \frac{2}{\Delta T} \arcsin\left(\min\left(\frac{\Delta T}{2} \frac{\|(\hat{\omega}_0 - \hat{\omega}_{-1})_\kappa\|}{\Delta T \|(\hat{\omega}_0)_\kappa\|}, 1\right)\right), \quad (53)$$

recovers $\sigma(\kappa)$ up to the FD-response bias that the arcsin removes: data-conditioning of the rate is structurally lossless.

Conditioned architecture (production: local taps). The deliverable keeps the zero-FFT locality of the estimator and conditions the stencils by *tap modulation*. The *taps* of a stencil are its weights, in the filter-theory sense: a width- W stencil is a discrete filter with W taps per direction. Conditioning modulates these taps as a function of the measured rate:

$$\text{taps}_d^{(k)} = \text{taps}_{d,\text{base}}^{(k)} + \left(\frac{\Delta T}{\Delta T_{\text{ref}}}\right)^{S-k} \Delta \text{taps}_{\theta,k,d}(\hat{\sigma}\text{-features}), \quad k \geq 1, \quad (54)$$

with the reference step ΔT_{ref} anchoring the exact scaling law at $\mathcal{O}(1)$ (the raw ΔT^{S-k} powers are far below unity across the sweep and would render the modulation path numerically dead), amplitude exactly zero on the $k = 0$ channels ($\varepsilon_0 = 0$: the undifferenced mark has no time error to absorb), and $\Delta \text{taps}_\theta$ a small shared MLP reading per-shell features of $x_i = \Delta T \hat{\sigma}(\kappa_i)$ at a fixed set of relative shells $\kappa_i/\kappa_{\text{max}}$. The output head is zero-initialized: at initialization the conditioned model is bit-identical to the unconditioned control. The production configuration:

quantity	value
reference step ΔT_{ref}	1.5×10^{-2}
modulation MLP	$10 \rightarrow 24 \rightarrow 24 \rightarrow 240$, tanh ($\approx 6.9\text{k}$ parameters)
per-shell features	$[x_i, \log(1+x_i)]$, $x_i = \Delta T \hat{\sigma}(\kappa_i)$
relative shells $\kappa_i/\kappa_{\text{max}}$	$\{0.15, 0.3, 0.5, 0.7, 0.85\}$
output head	zero-initialized

Empirical verification. The two-mark estimator (53) supplies the conditioning variable $\sigma(\kappa)$. Its consistency is assessed by the collapse of $r_j/\Delta T^{S-k}$ across the time-step sweep, as prescribed by Eq. (52).

6.4. Loss.

Offline derivative loss. Per-sample, per-order relative L_2 with equal weights over $[\dot{N}, \ddot{N}, \ddot{N}]$ (the error-propagation result of Sec. 4.2.4 makes equal weights correct), predictions 2/3-dealiased before the loss, and a norm floor $\max(\|t_c\|, 0.1 \times \text{member-median } \|t_c\|)$ in the denominator to cap the leverage of near-zero-target samples without dropping them. Medians are logged alongside floored means: relative errors on near-quiescent samples otherwise exert disproportionate influence on the optimizer (Sec. 8).

Rollout fine-tune. The offline objective does not see the closed loop, so the estimator is fine-tuned *through the stepper*. The supervised head is $\frac{1}{m} \sum_{j=1}^m \text{rel}L_2(\omega_j^{\text{ML}}, \omega_j^{\text{RK4}})$, where ω_j^{RK4} is obtained by accumulating fine-step RK4 updates over each coarse interval. Truncated backpropagation is applied over a window of w steps, while the rollout horizon m increases during training and is followed by a reference-free tail.

Reference-free tail regularization. The reference-free tail is regularized by two mechanisms:

1. *Analytic free-tail supervision.* On reference-free rolled steps the model heads are supervised against the on-the-fly *exact analytic* targets (5) of the pre-step rolled state (weight 10^{-2} , capped): the estimator must remain a consistent derivative estimator on its own trajectory, arbitrarily far from the data horizon.
2. *Stability regularization.* The principal and parasitic roots of Eq. (56) are penalized when they cross the unit disk over the resolved modal set.

7. Stability analysis of the learned correction. For a Fourier mode, the uncorrected AB2–CN2 amplification polynomial is

$$\pi_0(g) = g^2 - \left(r + \frac{3}{2}\tilde{\rho}\right)g + \frac{1}{2}\tilde{\rho}, \quad r = \frac{1 + z_L/2}{1 - z_L/2}, \quad \tilde{\rho} = \frac{z_N}{1 - z_L/2}. \quad (55)$$

Let $\hat{P}_p$ denote the implemented modal symbol of the order- p defect operator, including the finite-difference and convolution errors of the learned derivative slots and the retained RK4 target term. The corrected polynomial is

$$\pi_{\text{ADC}}(g) = \pi_0(g) + \sum_{p=3}^P \frac{\hat{P}_p(z_L, z_N; g)}{D_p} g. \quad (56)$$

The principal amplification factor is the root that continues from $g = 1$ at the origin, and stability requires every root to lie in the closed unit disk, with repeated unit-modulus roots excluded. Equation (56) places the implemented estimator error inside the polynomial coefficients; it does not replace that error by an external scalar perturbation. Writing $R_4(z) = 1 + z + z^2/2 + z^3/6 + z^4/24$, the principal root satisfies

$$g_{\text{ADC}}(z) = R_4(z/K)^K + \mathcal{O}(z^{P+1}) + \varepsilon_{\text{impl}}(z), \quad (57)$$

where $\varepsilon_{\text{impl}}$ contains the implemented reconstruction and mixed-precision errors. Thus, the finite-order correction follows the accumulated RK4 amplification factor through the assembled order, while the remaining AB2–CN2 tail and the estimator error determine the departure from the target. Figure 3 displays the associated scalar stability geometry.

For an S -level temporal reconstruction, the error in the m th nonlinear derivative slot satisfies

$$\hat{N}^{(m)} - N^{(m)} = C_m \Delta T^{S-m} \varepsilon_{\text{NN}} + \mathcal{O}(\Delta T^{S-m+1}), \quad m \geq 1, \quad (58)$$

where the exact implemented finite-difference symbol supplies the complete higher-order series. For a quadratic nonlinearity, the unrepresented cross terms are

$$N_{\text{quad}}^{(S)} = \sum_{j=0}^S \binom{S}{j} P(\omega^{(j)}, \omega^{(S-j)}). \quad (59)$$

The conservative stability certificate inserts the corresponding modal term with adversarial phase into every learned slot. Any component captured by the Wiener projection reduces this contribution, so the no-projection evaluation provides the limiting case for the selected local convolution class.

8. Mixed-precision implementation and comparative evaluation.

8.1. Arithmetic partition. Table 4 records the precision of every computational block.

Table 4: Mixed-precision partition of the corrected solver.

component	arithmetic
prognostic vorticity and stored nonlinear history	Float64
spectral transforms, linear symbols, and dealiasing	Float64
pseudo-spectral Jacobian and forcing	Float64
AB2–CN2 update and Crank–Nicolson solve	Float64
network weights, activations, and derivative predictions	Float32
defect coefficients, correction assembly, and state update	Float64

8.2. Comparison set. The corrected AB2–CN2 formulation is evaluated against the uncorrected AB2–CN2 method and higher-history Adams–Bashforth–Crank–Nicolson schemes. An end-to-end WENO configuration [19] is included to assess the tradeoff between the pseudo-spectral corrected solver and a nonlinear shock-capturing spatial discretization. Because WENO changes the spatial operator as well as the temporal work profile, this comparison is reported separately from the like-for-like temporal comparisons.

Table 5: Comparative configurations and controlled quantities.

configuration	spatial discretization	time integration	principal control
AB2–CN2	pseudo-spectral	AB2–CN2	base step
ADC–AB2–CN2	pseudo-spectral	corrected AB2–CN2	order and stencil depth
AB–CN baselines	pseudo-spectral	higher-history AB–CN	matched nonlinear work
RK4 target	pseudo-spectral	fine-step RK4	reference error
WENO baseline	WENO	AB2–CN2	matched grid error

The reported quantities are vorticity error, kinetic-energy and enstrophy error, spectral error over the resolved band, maximum stable step, nonlinear evaluations per simulated time unit, memory, and wall-clock time. Accuracy comparisons use a common high-resolution reference trajectory and identical physical parameters. Performance comparisons include data movement and precision-conversion costs, and all wall-clock values are obtained from synchronized repeated runs on the same hardware.

8.3. Training protocol. Training samples span forced and decaying quasi-geostrophic turbulence across multiple Reynolds numbers, β parameters, forcing scales, grid resolutions, and coarse time steps. Exact nonlinear-derivative targets are formed by the spectral chain rules of Sec. 2.2. The network is first optimized against the derivative targets and is then fine-tuned through corrected rollouts. The loss combines relative vorticity error, spectral error, conservation diagnostics, and the modewise stability penalty derived from Eq. (56).

9. Conclusions. We formulate a mixed-precision amortized defect correction that maps coarse AB2–CN2 updates to an accumulated fine-step RK4 trajectory for two-dimensional quasi-geostrophic vorticity flows. The construction retains the double-precision pseudo-spectral solver and confines Float32 arithmetic to a shallow estimator of nonlinear temporal derivatives. Because the correction

is expressed as the difference between the AB2–CN2 and RK4 local errors, the known Fourier-diagonal contributions remain analytical, while learning is restricted to the bilinear nonlinear-derivative terms. The finite-radius and finite-history error mechanisms, the exact propagation of the derivative reconstruction error, and the Wiener floor therefore remain explicit in the formulation. The corrected modewise amplification polynomial incorporates the implemented estimator error directly and provides a stability criterion for the complete numerical update. The comparative protocol separates Adams–Bashforth temporal baselines from the end-to-end WENO comparison and evaluates accuracy, stability, memory, and computational work under controlled conditions.

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